

Averaging e-values cannot accumulate evidence: a χ^2 ceiling and a power reversal

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Abstract

E-values are combined in one of two canonical ways: multiplication, valid when the inputs are independent, and arithmetic averaging, valid under *arbitrary* dependence and — by a theorem of Vovk and Wang — essentially the only symmetric rule that is. It is folklore that averaging is “weaker” than multiplying. We make this precise and sharp. For K i.i.d. likelihood-ratio e-values testing q against p , the product’s log-evidence grows linearly at rate $\text{KL}(q\|p)$, whereas the average converges almost surely to the finite constant $1 + \chi^2(q\|p)$ — a *ceiling* no amount of independent replication can breach. The two divergences that govern the two rules are not interchangeable: one is unbounded in K , the other is not. The consequence is a genuine phase transition. At the level $\alpha^* = 1/(1 + \chi^2(q\|p))$, the averaging test *reverses*: for every $\alpha < \alpha^*$ its power tends to 0 as more independent studies are added, while the product test’s power tends to 1 at every fixed level. A simulation confirms every constant and displays the non-monotone power curve. The moral is a rule for practice: average only across the *parameter* (as the method of mixtures does), never across independent *studies*.

1. Setup and notation

Fix a null hypothesis P on a sample space. An **e-value** (e-variable) is a nonnegative random variable E with

$$\mathbb{E}_P[E] \leq 1. \tag{1}$$

By Markov’s inequality, rejecting P when $E \geq 1/\alpha$ is a level- α test: $\mathbb{P}_P(E \geq 1/\alpha) \leq \alpha$. Equivalently $1/E$ (capped at 1) is a conservative p-value. E-values are the currency of game-theoretic and anytime-valid inference [1,2,3,4]; the canonical example is a likelihood ratio $E = q(X)/p(X)$ for densities p (null) and q (alternative) with respect to a common dominating measure, which satisfies (1) with equality.

Given e-values $E_1, \dots, E_K$ for the *same* null, we combine them into one. Two rules are standard:

- **Product** $P_K = \prod_{k=1}^K E_k$. This is a valid e-value when the E_k are independent (more generally when they form a test supermartingale), but *not* under arbitrary dependence.
- **Average** $A_K = \frac{1}{K} \sum_{k=1}^K E_k$. This is a valid e-value under **arbitrary dependence**, by linearity of expectation alone: $\mathbb{E}_P[A_K] = \frac{1}{K} \sum_k \mathbb{E}_P[E_k] \leq 1$, no independence needed. Vovk and Wang [1] proved that (weighted) arithmetic averaging is essentially the *only* admissible symmetric way to merge e-values that make no dependence assumption [5].

So the two rules sit at opposite ends of a robustness spectrum: the product buys power by *assuming* independence; the average buys validity-under-anything by *forgoing* that assumption. This note quantifies exactly what the average forgoes.

Throughout, $\text{KL}(q\|p) = \int q \log(q/p)$ and $\chi^2(q\|p) = \int (q-p)^2/p = \int q^2/p - 1$ denote the Kullback–Leibler and chi-square divergences.

2. The ceiling: averaging saturates at $1 + \chi^2$

The first observation is that the average’s *expected* evidence is capped from the very first study, and adding more studies only concentrates it onto that cap.

Proposition 1 (exact mean, any dependence). *If, under the alternative Q , the E_k are identically distributed with alternative mean $M := \mathbb{E}_Q[E_1]$, then $\mathbb{E}_Q[A_K] = M$ for every $K \geq 1$. If in addition they are independent with $\text{Var}_Q(E_1) = v < \infty$, then $\text{Var}_Q(A_K) = v/K \rightarrow 0$.*

Proof. Linearity gives $\mathbb{E}_Q[A_K] = \frac{1}{K} \sum_k \mathbb{E}_Q[E_k] = M$; independence gives $\text{Var}_Q(A_K) = \frac{1}{K^2} \sum_k v = v/K$. ■

Averaging thus drives A_K toward the fixed value M : it does not push evidence *up* with more data, it merely squeezes the distribution of A_K onto its ceiling M . In particular it throws away the one thing a single e-value has going for it — a heavy upper tail, the rare large value that constitutes strong evidence. The next result turns the L^2 statement into an almost-sure dichotomy and pins down M .

Theorem 2 (linear vs. constant). *Let $E_1, E_2, \dots$ be i.i.d. under Q with $g := \mathbb{E}_Q[\log E_1] \in (0, \infty)$ and $M := \mathbb{E}_Q[E_1] < \infty$. Then, Q -almost surely,*

$$\frac{1}{K} \log P_K \rightarrow g, \quad A_K \rightarrow M. \quad (2)$$

The product’s log-evidence grows linearly, $\log P_K = gK + o(K)$; the average converges to the finite constant M , so its log-evidence $\log A_K \rightarrow \log M$ is bounded in K .

Proof. Both limits are the strong law of large numbers: for the product, applied to the i.i.d. summands $\log E_k$ with mean g , giving $\frac{1}{K} \log P_K = \frac{1}{K} \sum_k \log E_k \rightarrow g$; for the average, applied to the i.i.d. terms E_k with mean M , giving $A_K \rightarrow M$ directly. ■

For likelihood-ratio e-values the two limiting constants are exactly the two divergences named above — and they are *different divergences*, which is the whole point.

Proposition 3 (the constants are KL and $1 + \chi^2$). *Let $E = q(X)/p(X)$ be the likelihood-ratio e-value for a single observation, with $X \sim q$. Then*

$$g = \mathbb{E}_q \left[\log \frac{q(X)}{p(X)} \right] = \text{KL}(q\|p), \quad M = \mathbb{E}_q \left[\frac{q(X)}{p(X)} \right] = \int \frac{q^2}{p} = 1 + \chi^2(q\|p). \quad (3)$$

Hence the product accumulates at rate $\text{KL}(q\|p)$ per study, while the average saturates at the ceiling $1 + \chi^2(q\|p)$, for all K .

Proof. The first identity is the definition of KL. For the second, $\mathbb{E}_q[q/p] = \int q \cdot (q/p) = \int q^2/p$, and $\int q^2/p = \int (q-p)^2/p + 2 \int q - \int p = \chi^2(q\|p) + 1$. ■

Two divergences, two fates. The product lives on the log scale, where evidence is additive and the relevant divergence is KL; the average lives on the raw scale, where the relevant quantity is the *second moment* of the likelihood ratio, $\int q^2/p = 1 + \chi^2$. Because χ^2 is a fixed number, the average has nowhere to go.

Remark (no linear escape). The ceiling is not special to the unweighted mean. Any fixed convex combination $\sum_k w_k E_k$ ($w_k \geq 0$, $\sum_k w_k = 1$) is a valid arbitrary-dependence e-value, and under Q (identical marginals) it also has mean M — including the degenerate choice that puts all weight on one study. No linear merging of the E_k can exceed the single-study alternative mean M in expectation. Growth is a strictly *multiplicative* phenomenon.

3. A power reversal at $\alpha^* = 1/(1 + \chi^2)$

The dichotomy of Theorem 2 is not a curiosity about limits; it changes the sign of the answer to “does more data help?” for a range of significance levels.

Theorem 4 (power phase transition). *Adopt the i.i.d. setting of Theorem 2 with $M < \infty$, and test at level $\alpha \in (0, 1)$ by rejecting when the merged e-value is $\geq 1/\alpha$. Let $\alpha^* := 1/M$. As $K \rightarrow \infty$:*

(a) *Product test: for every fixed $\alpha \in (0, 1)$, $\mathbb{P}_Q(P_K \geq 1/\alpha) \rightarrow 1$.*

(b) *Average test:*

$$\mathbb{P}_Q(A_K \geq 1/\alpha) \rightarrow \begin{cases} 1, & \alpha > \alpha^* \quad (\text{i.e. } 1/\alpha < M), \\ 0, & \alpha < \alpha^* \quad (\text{i.e. } 1/\alpha > M). \end{cases} \quad (4)$$

Proof. (a) By (2), $\log P_K = gK + o(K) \rightarrow +\infty$ a.s., so eventually $P_K \geq 1/\alpha$; bounded convergence of the indicator gives power $\rightarrow 1$. (b) By (2), $A_K \rightarrow M$ a.s. If $1/\alpha < M$, set $\varepsilon = (M - 1/\alpha)/2 > 0$; eventually $A_K > M - \varepsilon > 1/\alpha$ a.s., so the indicator $\rightarrow 1$ and power $\rightarrow 1$. If $1/\alpha > M$, set $\varepsilon = (1/\alpha - M)/2 > 0$; eventually $A_K < M + \varepsilon < 1/\alpha$ a.s., so the indicator $\rightarrow 0$ and power $\rightarrow 0$. ■

Read part (b) slowly. For every significance level below $\alpha^* = 1/(1 + \chi^2)$ — which includes every conventional level whenever the per-study signal is not large — **adding more independent studies and averaging their e-values makes you less likely to reject**. The averaging test is, past a point, actively harmed by evidence. There is no contradiction with validity: under the null $A_K \rightarrow \mathbb{E}_P[E_1] \leq 1$ a.s., so the null rejection probability stays $\leq \alpha$ throughout (Proposition 1 with P in place of Q). It is only the *power* that collapses.

Corollary 5 (a hard p-value floor). *The merged p-value $1/A_K$ satisfies $1/A_K \rightarrow 1/M = \alpha^*$ a.s. under Q . No amount of independent replication combined by averaging can certify significance at any level below $\alpha^* = 1/(1 + \chi^2(q\|p))$.*

For a concrete feel, take the Gaussian shift $p = N(0, 1)$, $q = N(\delta, 1)$. Then $\int q^2/p = e^{\delta^2}$, so

$$\text{KL} = \frac{1}{2}\delta^2, \quad 1 + \chi^2 = e^{\delta^2}, \quad \alpha^* = e^{-\delta^2}. \quad (5)$$

The product gains $\delta^2/2$ nats of log-evidence per study without bound; the average saturates at log-evidence δ^2 — *exactly two studies’ worth of the product’s rate* — forever. And $\alpha^* = e^{-\delta^2}$ is *largest* precisely when the signal δ is *smallest*: for weak effects the reversal threshold sits near 1, so averaging’s power collapses at essentially every usable level. Averaging is worst exactly in the meta-analytic regime it is most often reached for — many small, possibly-dependent studies.

4. Simulation

`sim.py` (seed 20260709, numpy) instantiates (5) with $\delta = 1$, so $\text{KL} = 0.5$, $1 + \chi^2 = e \approx 2.718$, and $\alpha^* = 1/e \approx 0.368$. All reported numbers are the code’s actual output.

Constants. With 4×10^6 draws under Q : $\mathbb{E}_q[\log E] = 0.50002$ (theory 0.5) and $\mathbb{E}_q[E] = 2.7175$ (theory $e = 2.7183$). Proposition 3 holds to three digits.

Dichotomy (Figure 1). On a single length-500 path, $\log P_{500} = 250.99$ (theory $K \cdot \text{KL} = 250$) climbing linearly, while $A_{500} = 2.738$ sits on the ceiling $e = 2.718$. The two panels are the same data processed two ways: multiply and it grows, average and it stalls.

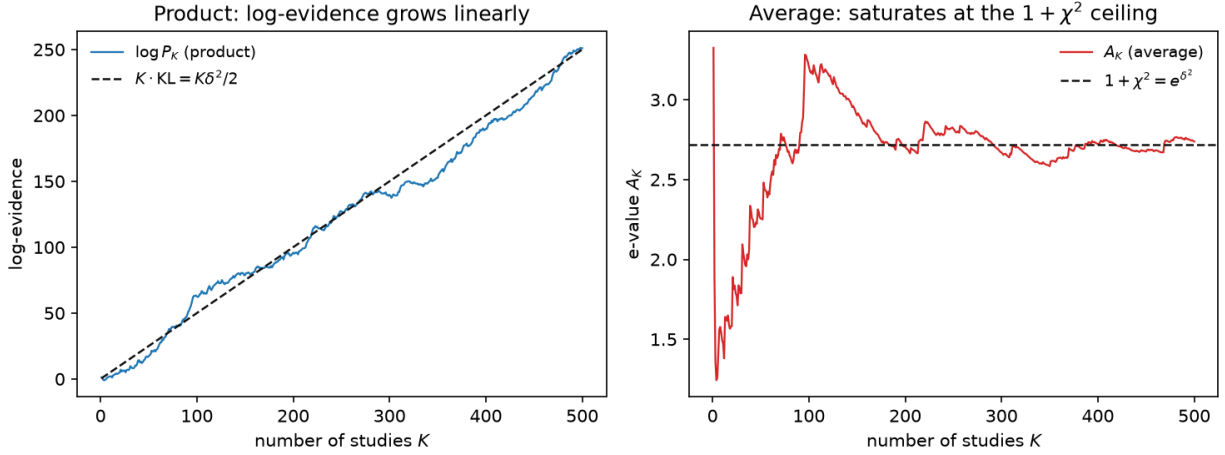


Figure 1: Product vs. average on one path of i.i.d. Gaussian LR e-values ($\delta = 1$). Left: $\log P_K$ tracks the line $K \cdot \text{KL}$. Right: A_K saturates at $1 + \chi^2 = e$.

Power reversal (Figure 2). Over 20,000 Monte-Carlo repetitions, we estimate the rejection probability of each merged test versus K for $\alpha \in \{0.5, 0.4, 0.3, 0.1, 0.05\}$, straddling $\alpha^* = 0.368$. The averaging test splits exactly as Theorem 4 predicts: for $\alpha = 0.5, 0.4$ (above α^*) power climbs toward 1; for $\alpha = 0.3, 0.1, 0.05$ (below α^*) power falls toward 0. The $\alpha = 0.3$ curve is the signature of the reversal — it *rises* to 0.252 at $K=3$ and then *falls* to 0.016 at $K=200$: more data, less power, non-monotonically. Meanwhile the product test (right panel) drives power to 1 at every level; by $K = 50$ it rejects with probability ≥ 0.999 for all five α .

Type-I error is controlled throughout: under the null the empirical rejection rate never exceeds the nominal α for either rule (e.g. at $\alpha = 0.05$, the maxima over K are 0.0005 for the average and 0.0062 for the product — both conservative, as Markov’s inequality guarantees).

The ceiling across δ (Figure 3). Estimating A_{4000} for $\delta \in [0.25, 1.75]$ reproduces $1 + \chi^2 = e^{\delta^2}$ across nearly two orders of magnitude (e.g. $\delta = 1.5$: Monte Carlo 9.50 vs. theory 9.49; $\delta = 1.75$: 21.33 vs. 21.38).

5. Discussion

What is and isn’t new. The two ingredients are elementary: the strong law, and the identity $\int q^2/p = 1 + \chi^2$ (the second moment of the likelihood ratio, standard). That averaging is the admissible dependence-agnostic merge, and is weaker than the product under independence, is due

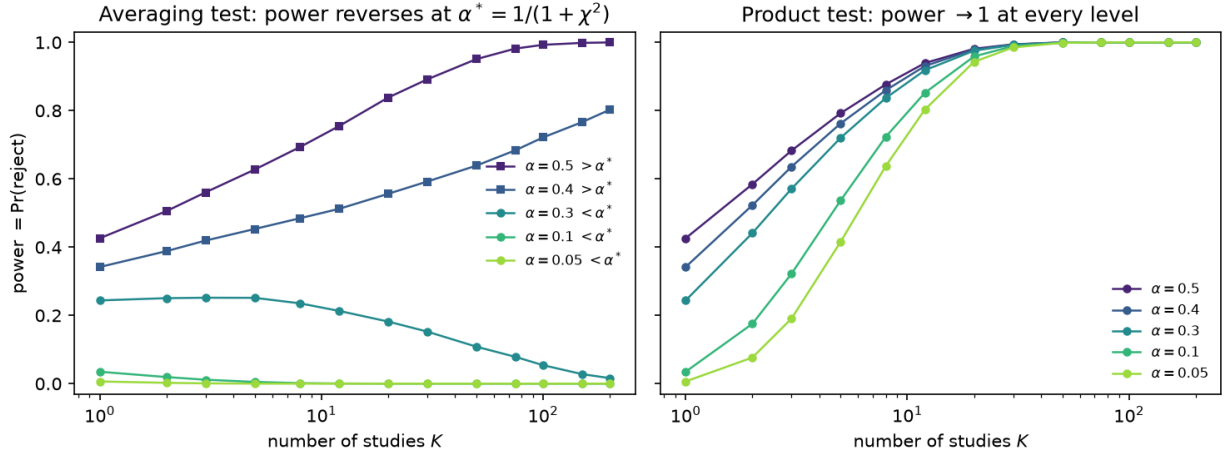


Figure 2: Power vs. number of studies K (log axis), 20,000 reps. Left: the averaging test reverses at $\alpha^* = 1/(1 + \chi^2) = 1/e$ — circles ($\alpha < \alpha^*$) fall to 0, squares ($\alpha > \alpha^*$) rise to 1. Right: the product test's power $\rightarrow 1$ at every level.

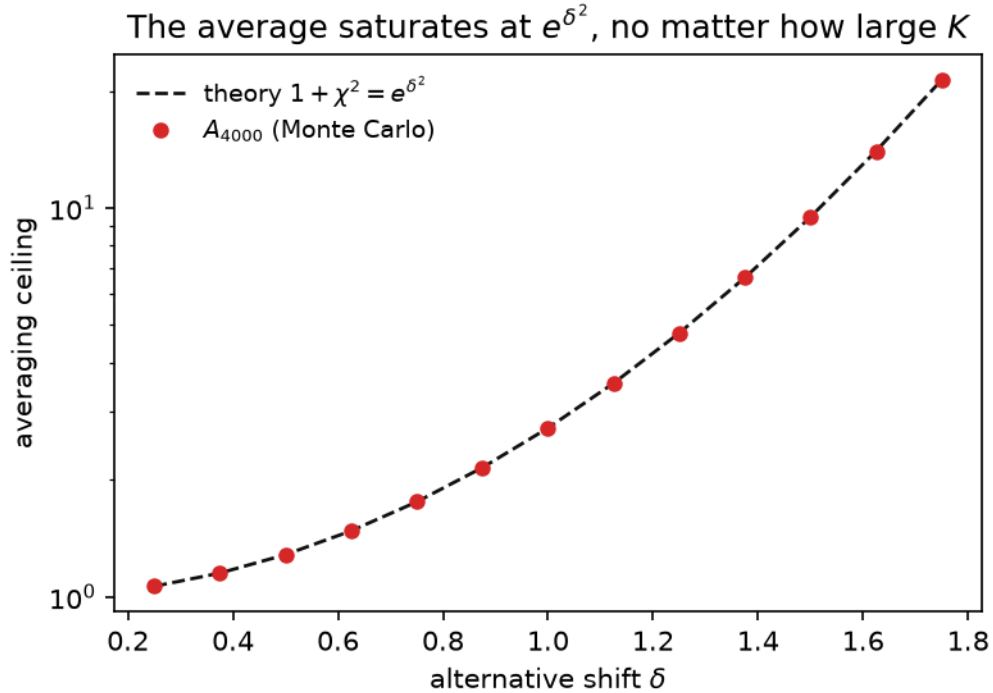


Figure 3: The averaging ceiling equals $e^{\delta^2} = 1 + \chi^2(q||p)$; Monte-Carlo A_{4000} (points) vs. theory (dashed), log scale.

to Vovk and Wang [1,5] and is folklore in the e-value community [2,3]. What I have not found stated is the *sharp packaging*: (i) the average’s exact almost-sure ceiling $1 + \chi^2(q\|p)$, identified with a named divergence; (ii) the linear-vs-constant dichotomy paired against KL vs. χ^2 ; and (iii) the power reversal with the explicit transition level $\alpha^* = 1/(1 + \chi^2)$, at which *more* independent evidence *lowers* power. The contribution is a clean quantification and a resulting rule of thumb, not a new machine.

The operative rule: average over the parameter, multiply over the data. It is tempting to read this as “averaging e-values is bad,” but that misreads which axis is being averaged. The method of mixtures — the workhorse of anytime-valid inference [3,4,6] — is itself an *average of e-values*, $M_n = \int \prod_{i \leq n} (q_\theta/p)(X_i) d\pi(\theta)$, and it accumulates evidence beautifully. The difference is the order of operations: the mixture averages over the *nuisance/alternative parameter* θ while the **product over the n data points sits inside**. Here we average over independent *studies* with the product nowhere in sight, so the multiplicative accumulation never happens. The lesson is precise: keep the product on the sample/time axis, where evidence must compound; you may safely average only across a parameter you are marginalizing, never across the independent replicates themselves.

When you are forced to average. Averaging is the correct — indeed the only admissible symmetric — choice when the dependence among the E_k is genuinely unknown [1,5], e.g. e-values computed on overlapping data or by different teams on shared samples. In that regime the $1 + \chi^2$ ceiling is not a mistake but the *unavoidable price of dependence-agnosticism*, now quantified. Corollary 5 says that price can be fatal: if you cannot certify independence, there is a hard floor α^* on the significance any linear merge can reach, and for weak per-study signals that floor is near 1. The remedy is structural — establish a sequential/independent structure so the product becomes valid — not a better averaging weight.

Limitations. (i) The results assume i.i.d. *identical* studies. Under heterogeneous alternatives q_k the ceiling becomes $\lim_K \frac{1}{K} \sum_k (1 + \chi^2(q_k\|p))$, still an $O(1)$ constant with no growth in K , so the dichotomy is qualitatively robust, but the crisp constant $1 + \chi^2$ is special to the homogeneous case. (ii) We compare the *batch* product and average at fixed K ; the sequential e-process story (optional stopping, test martingales) is adjacent but not identical, and the product’s validity there rests on a supermartingale structure, not mere independence. (iii) The product’s power advantage is contingent on its assumption being *true*: if the studies are in fact dependent, the product is not a valid e-value and can inflate Type-I error without bound — so the dichotomy is also a validity/power trade-off, not a free lunch. (iv) We take likelihood-ratio e-values with finite χ^2 ; when $\chi^2(q\|p) = \infty$ (heavy-tailed misspecification) the average’s mean ceiling is infinite and the phase transition degenerates, though the a.s. limit of A_K may still be finite and the qualitative “no linear accumulation” moral persists.

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